

$$7.3(a) \quad x(n) = R_N(n) \quad N=8$$

$$k \neq 0 \quad x(k) = \sum_{n=0}^7 e^{-j\frac{\pi}{4}kn} = \frac{1-e^{-j2\pi k}}{1-e^{-j\frac{\pi}{4}k}} = 0$$

$$k=0 \quad x(k) = 8$$

$$(c) \quad x(n) = e^{j\omega_0 n} R_N(n)$$

$$x(k) = \sum_{n=0}^7 e^{j(\omega_0 - \frac{\pi}{4}k)n} = \frac{1-e^{j(\omega_0 - \frac{\pi}{4}k)8}}{1-e^{j(\omega_0 - \frac{\pi}{4}k)}}$$

$$= \frac{1-e^{j(8\omega_0 - 2\pi k)}}{1-e^{j(\omega_0 - \frac{\pi}{4}k)}}$$

$$(e) \quad x(n) = n R_N(n)$$

$$x(k) = \sum_{n=0}^7 n e^{-j\frac{\pi}{4}kn} \quad e^{-j\frac{\pi}{4}kn} x(k) = \sum_{n=0}^7 n e^{-j\frac{\pi}{4}k(n+1)} = \sum_{n=1}^8 (n-1) e^{-j\frac{\pi}{4}kn}$$

$$x(k) - e^{-j\frac{\pi}{4}k} x(k) = \sum_{n=1}^7 e^{-j\frac{\pi}{4}kn} - 7e^{-j2\pi k}$$

$$= \frac{e^{-j\frac{\pi}{4}k} - 1}{1 - e^{-j\frac{\pi}{4}k}} - 7$$

$$\Rightarrow x(k) = \frac{-8}{1 - e^{-j\frac{\pi}{4}k}} \quad k \neq 0$$

$$x(k) = 28 \quad k=0$$

$$7.20 \quad f_1 = [-3, -1, 1, 3, 5, 3, 1, -1, -3, -5]$$

$$f_2 = [1, 2, 3, 4, 5, 3, 1, -1, -3, -5, -4, -3, -2, -1, 0, 0, 0, 0, 0]$$